GUI-LHADDAS

User Manual

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https://github.com/satoh-daiki/GUI-LHADDAS/

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References:

H. Nakayama, N. Onodera, D. Satoh, H. Nagai, Y. Hasegawa, Y. Idomura, “Development of local-scale high-resolution atmospheric dispersion and dose assessment system”, Journal of Nuclear Science and Technology, 2022.
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1. Introduction

1.1 What is GUI-LHADDAS?

GUI-LHADDAS is a graphical user interface (GUI) application for the Local-scale High-resolution Atmospheric Dispersion and Dose Assessment System (LHADDAS). LHADDAS is a coupled simulation system consisting of LOHDIM-LES, a large-eddy simulation (LES)-based atmospheric dispersion code, and SIBYL, a gamma-ray dose rate assessment code. It is designed to evaluate the transport of radioactive materials and the resulting dose rate distribution at the local scale (on the order of kilometers) with high spatial resolution (on the order of meters), explicitly resolving individual buildings.

GUI-LHADDAS provides an integrated environment for configuring simulation parameters, executing LOHDIM-LES and SIBYL, and visualizing the results in a three-dimensional geospatial environment. It eliminates the need for manual editing of text-based input files and manual transfer of output data between the two codes, making LHADDAS accessible to a broader range of users.

GUI-LHADDAS is a cross-platform desktop application built on the Electron framework and is available for Windows, macOS, and Linux. The source code is publicly available under the MIT License at https://github.com/satoh-daiki/GUI-LHADDAS, along with pre-built application packages for each supported platform.

1.2 System Requirements

GUI-LHADDAS has been verified to run on the following operating systems:

- Windows 11 (x64)
- macOS Sequoia 15.3 (Apple Silicon)
- Ubuntu 24.04 (x64)

1.3 Dependencies

GUI-LHADDAS requires LOHDIM-LES and SIBYL executables to perform atmospheric dispersion simulations and dose rate assessments, respectively. These codes must be obtained separately and placed in the appropriate directory before use (see Chapter 3).

LOHDIM-LES is distributed through the JAEA Program and Database Retrieval System (PRODAS) at https://prodas.jaea.go.jp. Access requires registration on the PRODAS website.

SIBYL is publicly available under the MIT License at https://github.com/satoh-daiki/SIBYL.

2. Installation

Pre-built application packages for each supported platform are available on the GitHub Releases page at https://github.com/satoh-daiki/GUI-LHADDAS/releases. Download the appropriate package for your platform and follow the instructions below.

2.1 Windows

Download `gui-lhaddas-1.1.0-win-setup.exe` and double-click it to launch the installer. The installation will proceed automatically. Once complete, a shortcut is created on the desktop and GUI-LHADDAS launches automatically.
The default installation directory is:

```
C:/Users/<username>/AppData/Local/Programs/gui-lhaddas
```

Note: Windows may display a security warning (Windows Defender SmartScreen) during installation. If this occurs, click **More info** and then **Run anyway** to proceed.

2.2 macOS

Download `gui-lhaddas-1.1.0-mac.dmg` and double-click it to open the disk image. Drag the GUI-LHADDAS application to your Applications folder (or any preferred location). Double-click the application to launch it.

2.3 Linux

Download `gui-lhaddas-1.1.0-linux.AppImage`. Before launching, grant execute permission by running the following command in a terminal:

```
bash
chmod +x gui-lhaddas-1.1.0-linux.AppImage
```

Then double-click the file in your file manager to launch GUI-LHADDAS.

Note (Ubuntu 24.04): Double-clicking the AppImage on the desktop may not launch the application. Open the containing folder in the file manager and double-click the file from there.

Note: If an error related to fuse is displayed, install `libfuse2` with the following command and try again:

```
bash
sudo apt install libfuse2
```

3. Getting Started

This chapter walks through a complete simulation workflow using the Oklahoma City example case. Unlike the application itself, the example files are not bundled with the source code; they are distributed separately via the GitHub Releases page. Download the example archive (`examples.zip`) from Releases and extract it so that the Oklahoma City example is located at:

`GUI-LHADDAS/examples/OklahomaCity`
(see `examples/README.md` in the repository for download details)

3.1 Launching GUI-LHADDAS

Launch GUI-LHADDAS by double-clicking the desktop shortcut (Windows) or the application icon (macOS, Linux). The main window appears as shown in **Fig. 3-1**, consisting of three panels:

- **(A) Map Panel**: displays the three-dimensional geospatial environment and simulation results.
- **(B) Map Control Panel**: provides controls for the release point, map source, map alignment, and interface language.
- **(C) Simulation Control Panel**: contains five tabs for configuring and executing simulations.

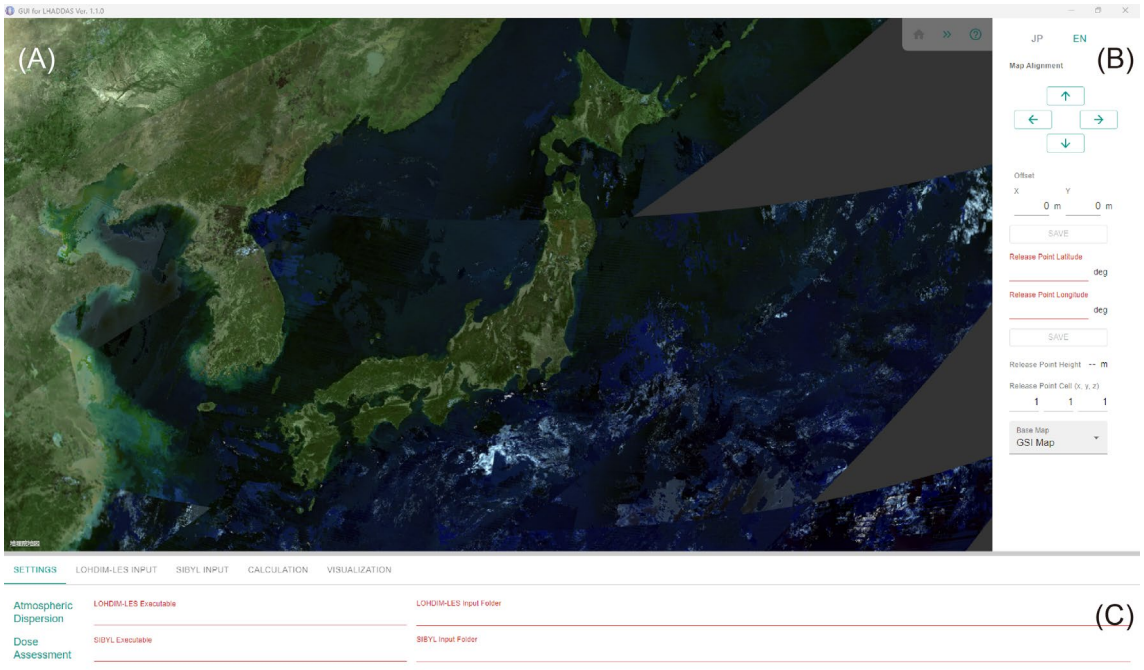


Fig. 3-1. Main window of GUI-LHADDAS showing the three-panel layout: (A) Map Panel, (B) Map Control Panel, and (C) Simulation Control Panel.

3.2 Initial Setup (Map Control Panel and Settings Tab)

This example uses the Oklahoma City, USA, as the target domain. Since the default map source (GSI Maps) covers Japan only, the map source must be switched to OpenStreetMap before configuring the simulation paths. To switch the map source, locate the map source selector in the Map Control Panel and select **OpenStreetMap**.

Next, select the **Settings** tab in the Simulation Control Panel and configure the following paths as shown in **Fig. 3-2**:

- **LOHDIM-LES Executable**: the path to the LOHDIM-LES executable file.
- **SIBYL Executable**: the path to the SIBYL executable file.
- **LOHDIM-LES Input Folder**: the directory for LOHDIM-LES input and output files.
- **SIBYL Input Folder**: the directory for SIBYL input and output files.

The I/O directories for LOHDIM-LES and SIBYL can be set to the same directory or to separate directories. For the Oklahoma City example, set both I/O directories to:

```
GUI-LHADDAS/examples/OklahomaCity
```

Once the I/O directories are specified, GUI-LHADDAS automatically loads any existing input files found in those directories. Since `SourceLocation.txt` is present in `GUI-LHADDAS/examples/OklahomaCity`, the Map Panel automatically navigates to the Oklahoma City area and renders the release point, simulation domain boundary, building geometry, and computational grid. The format and manual configuration of `SourceLocation.txt` are described in Chapter 4.

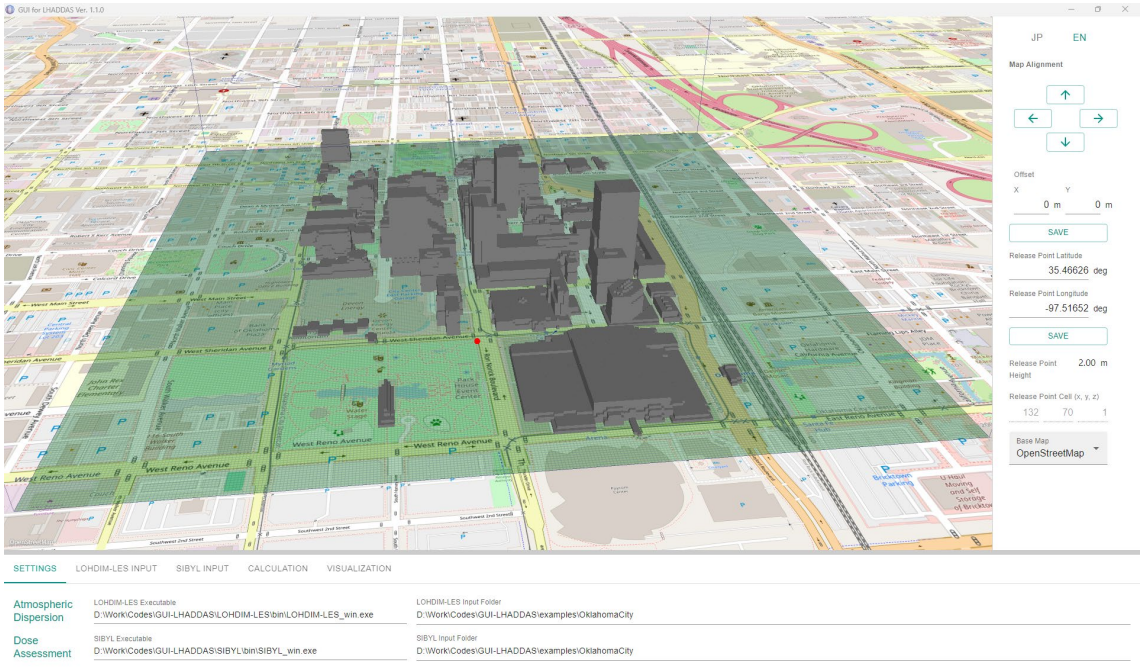


Fig. 3-2. Main window of GUI-LHADDAS after loading the Oklahoma City example, showing the configured paths in the Settings tab and the rendered simulation domain in the Map Panel.

3.3 Reviewing LOHDIM-LES Input Parameters (LOHDIM-LES Input Tab)

Select the **LOHDIM-LES Input** tab. Because the Oklahoma City input files are already present in the specified I/O directory, the simulation parameters are loaded automatically, as shown in **Fig. 3-3**. Review the parameters to confirm they are correct. No manual saving or reloading is required for this example.

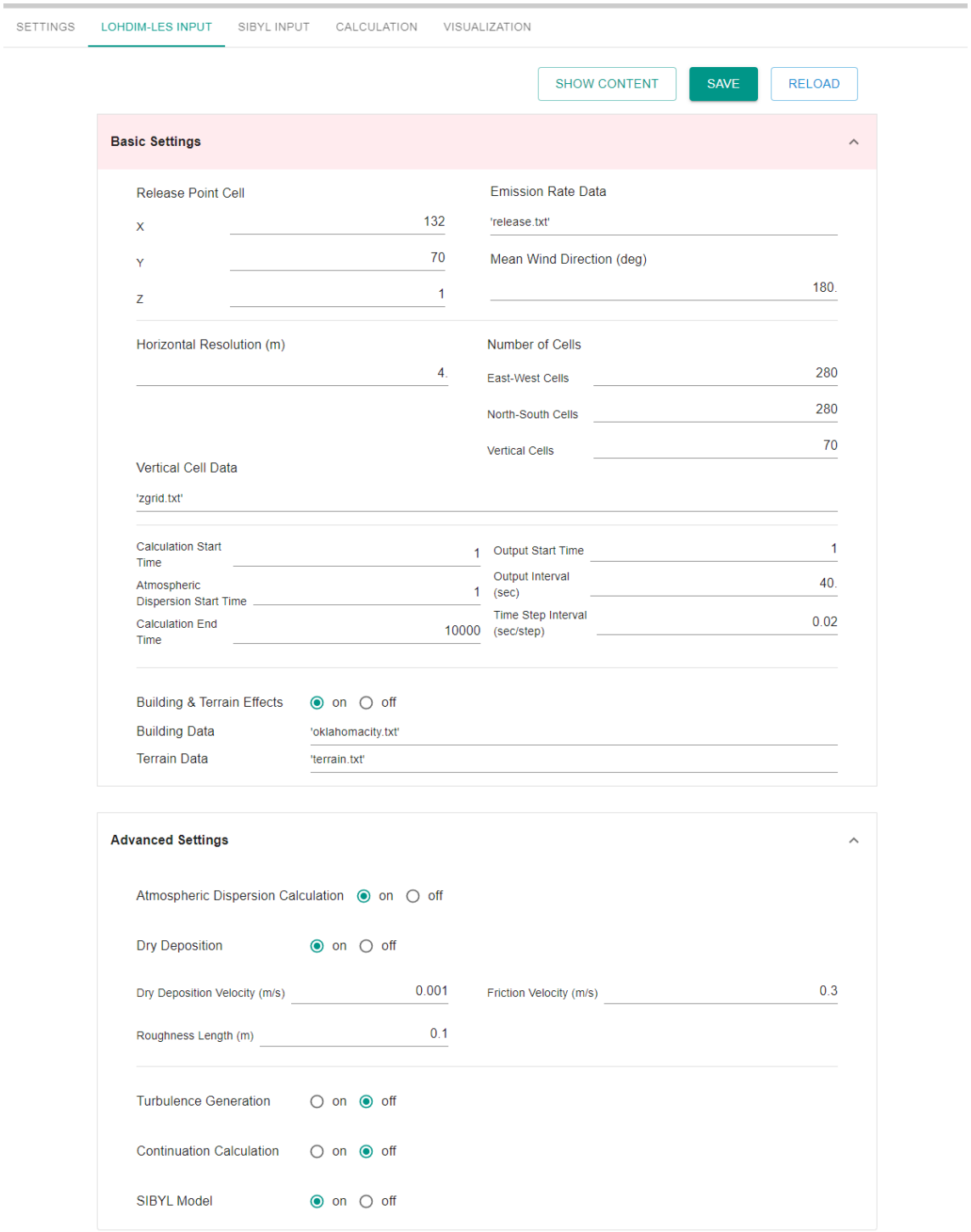


Fig. 3-3. LOHDIM-LES Input tab with parameters automatically loaded from the Oklahoma City example files.

3.4 Reviewing SIBYL Input Parameters (SIBYL Input Tab)

Select the **SIBYL Input** tab. As with LOHDIM-LES, the SIBYL parameters are loaded

automatically from the specified I/O directory, as shown in **Fig. 3-4**. Review the parameters to confirm they are correct.

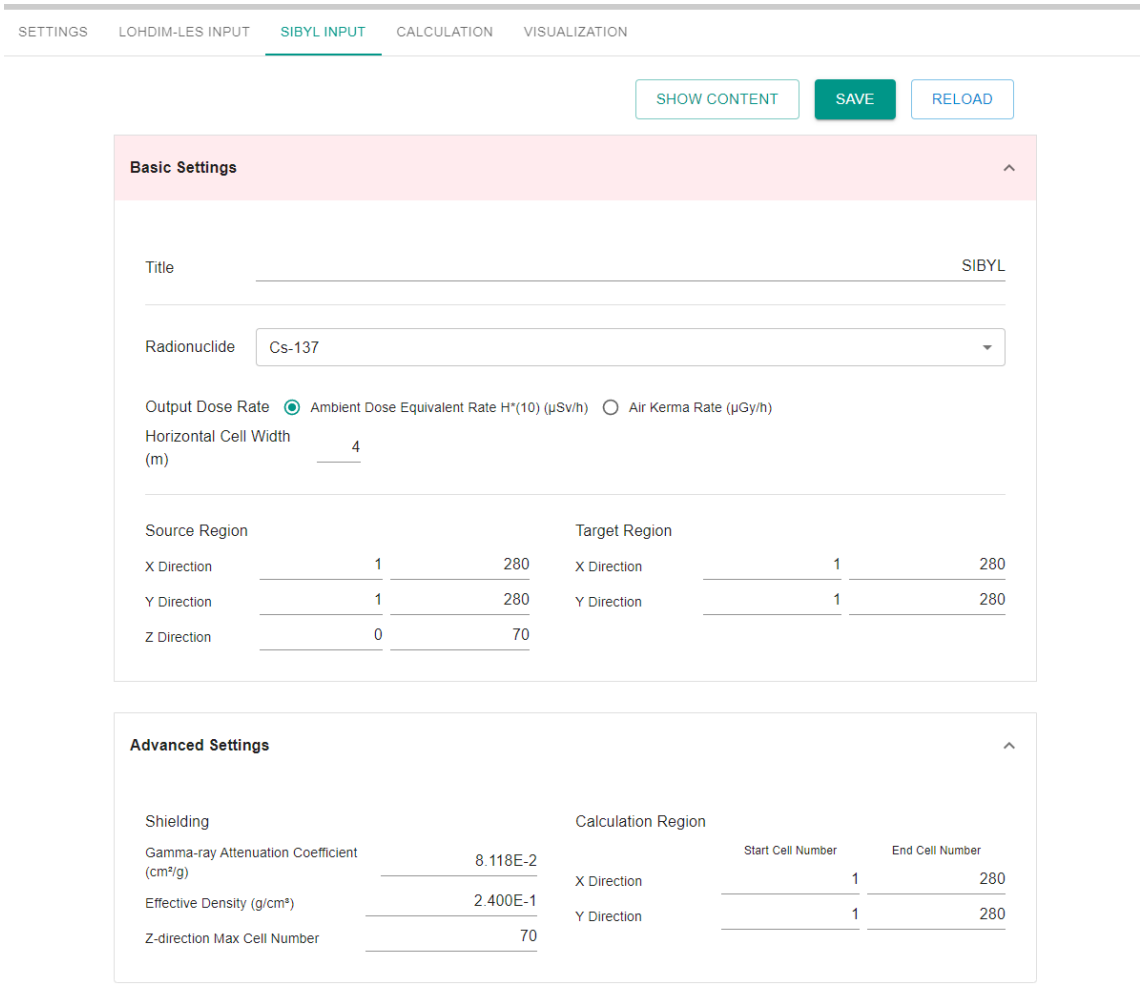


Fig. 3-4. SIBYL Input tab with parameters automatically loaded from the Oklahoma City example files.

3.5 Running the Simulation (Calculation Tab)

Select the **Calculation** tab, as shown in **Fig. 3-5**. Both LOHDIM-LES and SIBYL are enabled by default. Click **START CALCULATION** to begin the simulation.

The simulation proceeds as follows:

1. LOHDIM-LES launches in a terminal window and begins the atmospheric dispersion calculation.
2. When LOHDIM-LES completes, the terminal closes and SIBYL launches automatically

in a new terminal window.

3. When SIBYL completes, the terminal closes and the progress bar is fully filled.

Note: Ensure that the computer does not enter sleep mode during the calculation, as this will interrupt the simulation.

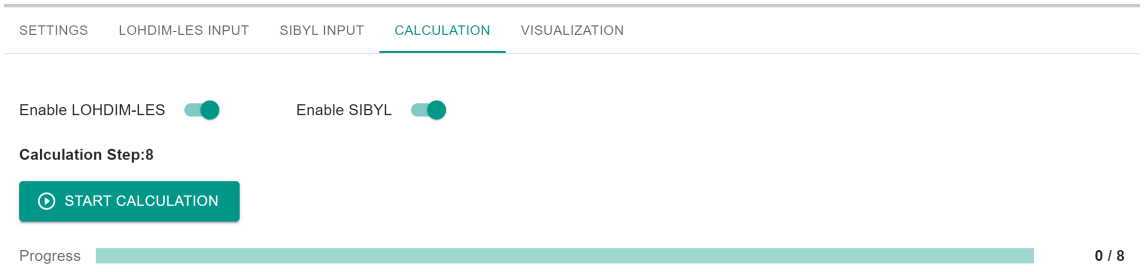


Fig. 3-5. Calculation tab during simulation execution.

3.6 Visualizing the Results (Visualization Tab)

When the simulation is complete, select the **Visualization** tab to display the results. **Fig. 3-6** shows the three-dimensional plume dispersion computed by LOHDIM-LES, and **Fig. 3-7** shows the ground-level dose rate distribution computed by SIBYL. Detailed instructions for controlling the visualization are provided in Chapter 5.

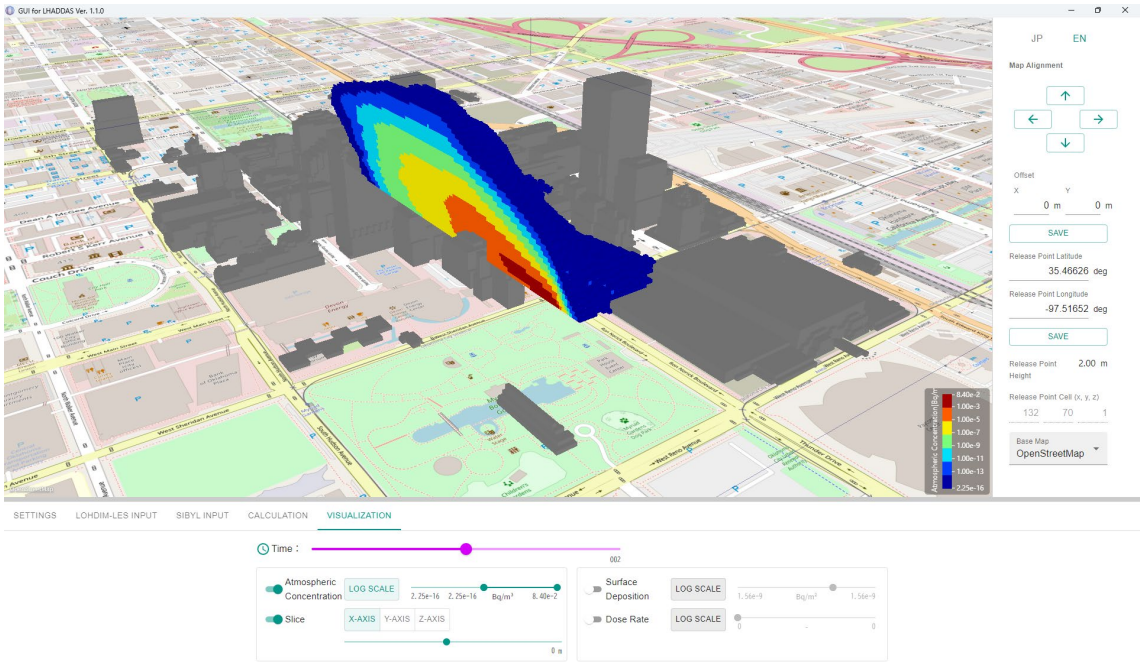


Fig. 3-6. Three-dimensional visualization of plume dispersion through the Oklahoma City street

network (LOHDIM-LES result).

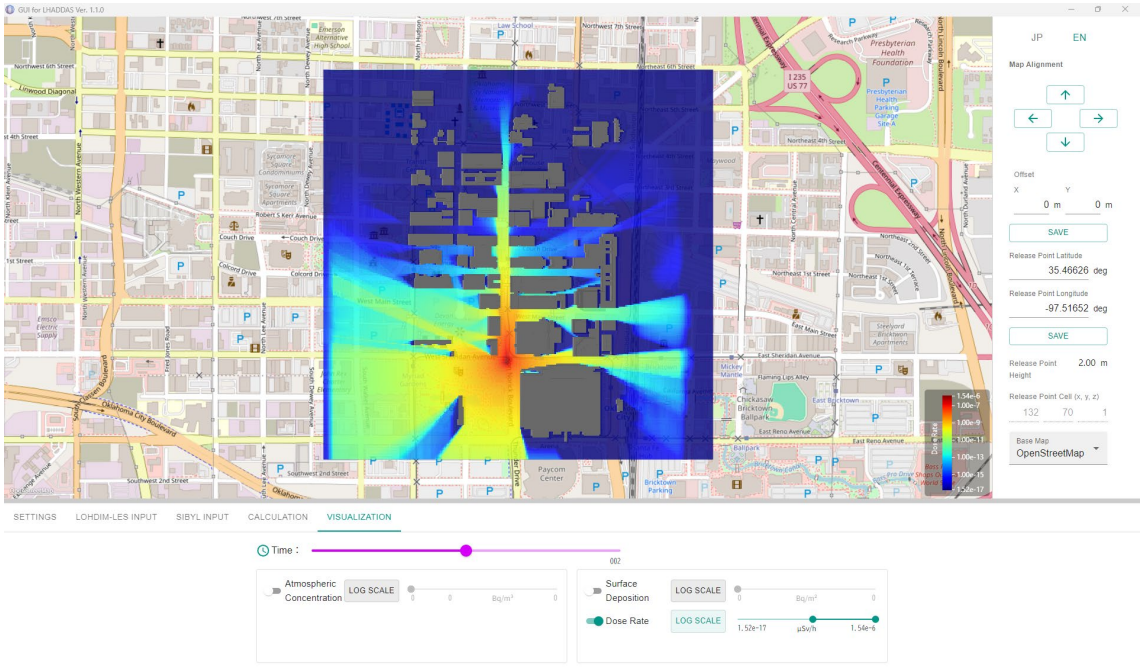


Fig. 3-7. Ground-level dose rate distribution over the Oklahoma City domain (SIBYL result).

4. Map Panel and Map Control Panel

4.1 Map Navigation

The Map Panel (A) and Map Control Panel (B) are shown in **Fig. 4-1**. The Map Panel displays a three-dimensional geospatial environment. When input data are loaded, the simulation domain boundary and computational grid are rendered, with buildings shown in gray.
The following mouse operations are available for navigating the Map Panel:

Operation	**Mouse action**
Pan	Drag
Rotate (ground-based)	Middle button drag, or Ctrl + drag
Rotate (viewpoint-based)	Shift + drag
Zoom	Mouse wheel, or right button drag

Clicking the **Home** icon in the upper right corner of the Map Panel resets the view to the default position and orientation.

Setting the release point: Double-clicking a location on the map sets it as the release point, indicated by a red marker. The coordinates are automatically reflected in the **Release Point Latitude** and **Release Point Longitude** fields in the Map Control Panel. Alternatively, coordinates can be entered directly in these fields. Clicking **SAVE** in the Map Control Panel saves the coordinates to `SourceLocation.txt` in the I/O directory.

If `SourceLocation.txt` already exists in the I/O directory when the directory is specified, the coordinates are loaded automatically, and the Map Panel navigates to the corresponding location. The file format is as follows:

```
(Latitude,Longitude)
35.46626,-97.51652
```

The first line is a fixed header. The second line contains the latitude and longitude in decimal degrees, separated by a comma. Positive values indicate north latitude and east longitude; negative values indicate south latitude and west longitude.

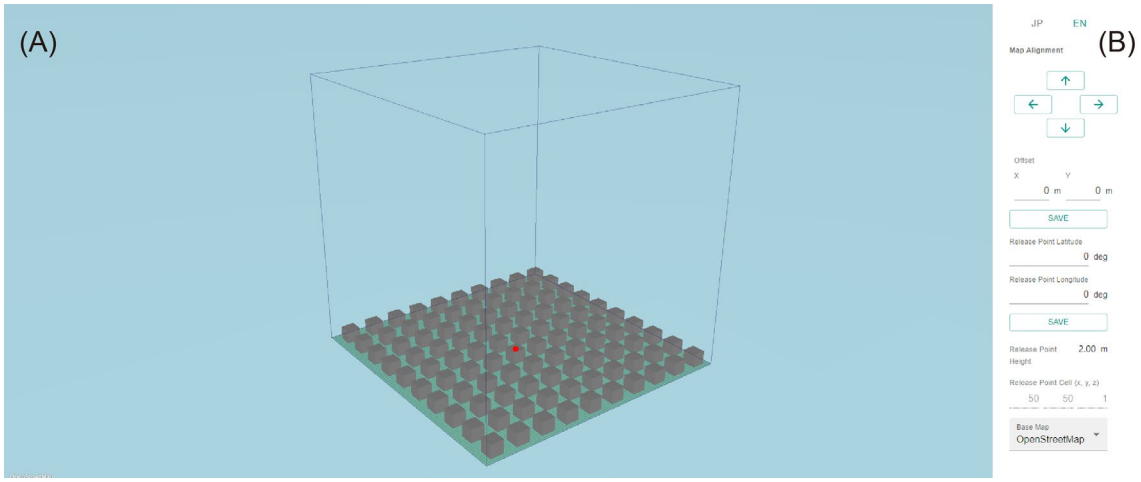


Fig. 4-1. Map Panel (A) and Map Control Panel (B).

4.2 Map Source Selection

GUI-LHADDAS supports two map sources: **GSI Maps**, provided by the Geospatial Information Authority of Japan, and **OpenStreetMap**. The map source can be selected from the **Base Map** dropdown menu in the Map Control Panel.

When the map tile servers cannot be reached, for example when the computer is offline, the map is not displayed, but all other functions of GUI-LHADDAS, including the calculation and the visualization of the results, work normally.

Note: GSI Maps covers Japan only. Select OpenStreetMap when working with domains outside Japan.

4.3 Language Switching

The interface language can be switched between Japanese and English using the **JP** / **EN** toggle at the top of the Map Control Panel.

4.4 Map Alignment

Depending on the data source used to construct the computational domain, a slight misalignment may exist between the map tile imagery and the rendered building geometry. While this does not affect simulation results, it can appear untidy when visualizing results overlaid on the map. The Map Alignment controls in the Map Control Panel allow the rendering position of the computational domain to be corrected.

The offset can be adjusted in two ways. Clicking the arrow buttons (north, south, east,

west) shifts the domain by 10 m per click in the corresponding direction; the resulting offset values are reflected in the **Offset X** and **Offset Y** fields. Alternatively, offset values can be entered directly in these fields without restriction. Clicking **SAVE** saves the offset to `MapCorrection.txt` in the I/O directory. On subsequent launches, if `MapCorrection.txt` is found in the I/O directory, the offset correction is applied automatically.

5. Simulation Control Panel

5.1 Settings tab

The Settings tab is used to configure the paths required before running a simulation. The tab is divided into two sections: **Atmospheric Dispersion** and **Dose Assessment**.

In the **Atmospheric Dispersion** section, specify the following:

- **LOHDIM-LES Executable**: the full path to the LOHDIM-LES executable file. LOHDIM-LES can be placed in any directory on the system.
- **LOHDIM-LES Input Folder**: the directory in which LOHDIM-LES input and output files are stored.

In the **Dose Assessment** section, specify the following:

- **SIBYL Executable**: the full path to the SIBYL executable file. SIBYL can be placed in any directory on the system.
- **SIBYL Input Folder**: the directory in which SIBYL input and output files are stored.

Paths can be specified by hovering over the input field to reveal a folder icon, then clicking it to open the system file manager (File Explorer on Windows, Finder on macOS, or the equivalent file manager on Linux).

The LOHDIM-LES Input Folder and SIBYL Input Folder can be set to the same directory or to separate directories. Once a folder path is specified, GUI-LHADDAS automatically scans the directory and loads any existing input files. If `SourceLocation.txt` or `MapCorrection.txt` are present, they are also loaded automatically (see Chapter 4).

5.2 LOHDIM-LES Input tab

The LOHDIM-LES Input tab allows users to configure the input parameters for LOHDIM-LES. The available parameters and their default values are listed in **Table 5-1**.

Table 5-1. Input parameters of the LOHDIM-LES Input tab and their default values.

Basic Settings	Explanation
Release Point Cell X	(integer) Cell number of the release point along the x axis within the computational domain. Default: 50
Y	(integer) Cell number of the release point along the y axis within the computational domain. Default: 20

Z	(integer) Cell number of the release point along the z axis within the computational domain. Default: 1
Emission Rate Data	(character) Name of the file containing the emission rate data. Note: the filename must be enclosed in single quotes. Default: 'release.txt'
Mean Wind Direction	(float) Mean wind direction. Unit: degrees Default: 180
Horizontal Resolution	(float) Horizontal resolution of the computational cells. Unit: m Default: 4
Number of Cells East-West Cells	(integer) Number of cells in the east-west direction (x axis) of the computational domain. Default: 100
North-South Cells	(integer) Number of cells in the north-south direction (y axis) of the computational domain. Default: 100
Vertical Cells	(integer) Number of cells in the vertical direction (z axis) of the computational domain. Default: 68
Vertical Cell Data	(character) Name of the file containing the vertical (z axis) grid spacing data. Note: the filename must be enclosed in single quotes. Default: 'zgrid.txt'
Calculation Start Time	(integer) Time step number at which the calculation starts. Unit: step Default: 1

Atmospheric Dispersion Start Time	(integer) Time step number at which the atmospheric dispersion calculation starts, counted from the beginning of the simulation. Unit: step Default: 1
Calculation End Time	(integer) Time step number at which the calculation ends. Unit: step Default: 12000
Output Start Time	(integer) Time step number at which data writing to the output file begins. Unit: step Default: 6001
Output Interval	(float) Time interval at which results are written to the output file. With the default values of this parameter (60 s) and Time Step Interval (0.02 s/step), the atmospheric dispersion results are averaged and written to the file every 3000 time steps. Unit: s Default: 60
Time Step Interval	(float) Duration of each computational time step. Unit: s/step Default: 0.02
Building & Terrain Effects	(radio button) Enables or disables loading of building and terrain data. Default: on
Building Data	(character) Name of the file containing the building data. Note: the filename must be enclosed in single quotes. Default: ‘building.txt’
Terrain Data	(character) Name of the file containing the terrain data. Note: the filename

	must be enclosed in single quotes. Default: ‘terrain.txt’
Advanced Settings	Explanation
Atmospheric Dispersion Calculation	(radio button) Enables or disables the atmospheric dispersion calculation. Default: on
Dry Deposition	(radio button) Enables or disables the dry deposition calculation onto the ground surface. Default: off
Dry Deposition Velocity	(float) Dry deposition velocity onto the ground surface. Unit: m/s Default: 0.001
Friction Velocity	(float) Friction velocity of the ground surface. Unit: m/s Default: 0.3
Roughness Length	(float) Roughness length of the ground surface. Unit: m Default: 0.1
Turbulence Generation	(radio button) Enables or disables turbulence generation. Default: off
Continuation Calculation	(radio button) Enables or disables continuation calculation. Default: off
SIBYL Model	(radio button) Enables or disables the SIBYL dose rate distribution calculation after the atmospheric dispersion calculation. Default: on

The following files are required to run LOHDIM-LES: `LOHDIM-LES_input.data`, which is the main input file generated by GUI-LHADDAS from the parameters configured in this tab; `release.txt`, which contains the emission rate data (the filename can be changed via the

Emission Rate Data parameter); and `zgrid.txt`, which specifies the vertical grid spacing (the filename can be changed via the Vertical Cell Data parameter). If **Building & Terrain Effects** is enabled, `building.txt` and `terrain.txt` are also required, and their filenames can be changed via the **Building Data** and **Terrain Data** parameters, respectively.

If `LOHDIM-LES_input.data` exists in the LOHDIM-LES Input Folder specified in the Settings tab, the parameters are loaded automatically when the folder is specified. If the file is not found, the input fields are populated with the default values shown in **Table 5-1**. If `LOHDIM-LES_input.data` has been edited directly outside GUI-LHADDAS, click **RELOAD** to reload the file and reflect the updated values in the input fields.

Clicking **SHOW CONTENT** displays the current parameters formatted as a `LOHDIM-LES_input.data` file, allowing users to verify the content before saving. Once confirmed, click **SAVE** to write the parameters to `LOHDIM-LES_input.data` in the I/O directory. Navigating to another tab before clicking **SAVE** will discard any unsaved changes, and parameters edited in the GUI are not reflected in the calculation until **SAVE** is clicked. GUI-LHADDAS performs type checking on each input field, checks that the release point lies within the computational domain, and displays a warning if an invalid value is entered.

5.3 SIBYL Input tab

The SIBYL Input tab allows users to configure the input parameters for SIBYL. The available parameters and their default values are listed in **Table 5-2**. For a detailed description of SIBYL and its parameters, refer to the SIBYL user manual at https://github.com/satoh-daiki/SIBYL/blob/main/SIBYL-manual.pdf.

Table 5-2. Input parameters of the SIBYL Input tab and their default values.

Basic Settings	Explanation
Title	(character) Title of the simulation. Default: SIBYL
Radionuclide	(drop down) Radionuclide to be used as the gamma-ray source for dose rate distribution calculation. Default: Cs-137
Output Dose Rate	(radio button) Output dose rate quantity. Select either the ambient dose equivalent rate H*(10) or air kerma rate.

	Unit: μSv/h or μGy/h Default: H*(10)
Horizontal Cell Width	(integer) Horizontal (x-y plane) resolution of the computational grid. Unit: m Default: 4
Source Region X Direction	(integer) Start and end cell numbers of the source region along the x axis. Default: (1, 100)
Y Direction	(integer) Start and end cell numbers of the source region along the y axis. Default: (1, 100)
Z Direction	(integer) Start and end cell numbers of the source region along the z axis. Default: (0, 68)
Target Region X Direction	(integer) Start and end cell numbers of the target region along the x axis. The values must be within the source region. Default: (1, 100)
Y Direction	(integer) Start and end cell numbers of the target region along the y axis. Default: (1, 100)
Advanced Settings	Explanation
Shielding Gamma-ray Attenuation Coefficient	(float) Total attenuation coefficient of obstacles such as buildings. Unit: cm^2/g Default: 8.118E-2
Effective Density	(float) Effective density of obstacles. Unit: g/cm^3 Default: 2.400E-1
Z-direction Max Cell Number	(integer) Maximum cell number containing obstacles along the z axis.

	SIBYL skips the obstacle search in the region above the cell specified by this parameter, which can reduce the computational cost. Default: 68
Calculation Region X Direction	(integer) Start and end cell numbers of the dose-calculation region along the x axis. By default, the dose-calculation region corresponds to the target region, and obstacle data are provided for this region via `SHIELD.data`. These parameters allow users to narrow the calculation region without modifying `SHIELD.data`, which can be useful for test calculations. Default: (1, 100)
Y Direction	(integer) Start and end cell numbers of the dose-calculation region along the y axis. Default: (1, 100)

The following files are required to run SIBYL: `SIBYL_input.data`, which contains the parameters configured in this tab; `Z.data`, which specifies the vertical grid spacing; and `SIBYL_PLUME_NNN.data` and `SIBYL_GROUND_NNN.data`, which contain the air concentration and surface deposition at each output time (see Chapter 6). If building shielding is to be considered, `SHIELD.data` is additionally required; if terrain elevation is to be considered, `ELEVATION.data` is also required.

When LOHDIM-LES is enabled and SIBYL is executed in sequence following LOHDIM-LES, all of these files (`SIBYL_input.data`, `Z.data`, `SHIELD.data`, `ELEVATION.data`, `SIBYL_PLUME_NNN.data`, and `SIBYL_GROUND_NNN.data`) are generated automatically from the LOHDIM-LES output. In this case, manual preparation of these files is not required.

The behavior of the **SHOW CONTENT**, **SAVE**, and **RELOAD** buttons, as well as the automatic loading of parameters when `SIBYL_input.data` is found in the SIBYL Input Folder, are identical to those described for the LOHDIM-LES Input tab in Section 5.2.

GUI-LHADDAS performs type and range checking on each input field and displays a warning if an invalid value is entered.

5.4 Calculation tab

The Calculation tab controls the execution of LOHDIM-LES and SIBYL. **Fig. 3-5**

shows the tab as it appears before a calculation is started.

The **Enable LOHDIM-LES** and **Enable SIBYL** toggles allow each code to be independently enabled or disabled. When both codes are enabled, LOHDIM-LES runs first, and SIBYL starts automatically upon its completion, using the output files generated by LOHDIM-LES as input (see Section 5.3). When only SIBYL is enabled, the required SIBYL input files (`SIBYL_input.data`, `Z.data`, `SHIELD.data`, `ELEVATION.data`, `SIBYL_PLUME_NNN.data`, and `SIBYL_GROUND_NNN.data`) must be prepared manually in the SIBYL Input Folder before starting the calculation.

Calculation Step displays the total number of output time steps to be produced by the enabled code(s), as determined from the input file(s).

Click **START CALCULATION** to begin the simulation. If the button is grayed out, a required file has not been found. The button is enabled when the executable file and the I/O directory of each enabled code are specified in the Settings tab (see Section 5.1) and, when LOHDIM-LES is enabled, `LOHDIM-LES_input.data` and, if buildings and terrain are considered, the building and terrain files are found in the LOHDIM-LES I/O directory. The other input files are not checked at this stage; if one of them is missing, the code stops when it is run (see below).

Once the calculation starts, the executable for each enabled code is launched in a terminal window. The terminal closes automatically when the code completes. When both codes are enabled, the SIBYL terminal opens after the LOHDIM-LES terminal has closed. The progress bar advances each time an output file is written, at the interval specified by the **Output Interval** parameter in the LOHDIM-LES Input tab (see Section 5.2).

If an error occurs during a calculation, it is handled by LOHDIM-LES or SIBYL itself, and the messages of the code are shown in its terminal window. GUI-LHADDAS does not display an error message; the Calculation tab simply returns to its state before the calculation. Because the terminal window closes when the code stops, a failed run is recognized from the output files missing in the I/O directory. In the current version, the messages of LOHDIM-LES and SIBYL are not saved to log files; this will be added in the next update.

When both codes are enabled, GUI-LHADDAS checks, after LOHDIM-LES has finished, that its output files have been written; if they have not, SIBYL is not started. Whether it is run after LOHDIM-LES or alone, SIBYL is run once for each output time for which `SIBYL_PLUME_NNN.data` or `SIBYL_GROUND_NNN.data` is found in the SIBYL I/O directory (see Chapter 6), and if SIBYL produces no result for one output time, the remaining output times are skipped.

LOHDIM-LES saves its state at each output time, and GUI-LHADDAS stores it as `restart.dat` in the LOHDIM-LES I/O directory. A LOHDIM-LES calculation that has stopped can therefore be continued from the last output time by enabling **Continuation** in the LOHDIM-

LES Input tab (see Table 5-1).

5.5 Visualization tab

The Visualization tab controls the rendering of simulation results in the Map Panel. The **Time** slider at the top of the tab selects the output time step to display. Dragging the slider or clicking the desired position updates the Map Panel rendering accordingly; the current step index is shown to the right of the slider. Alternatively, clicking the label next to the slider reveals a text box in which a step index can be entered directly. Invalid values are ignored.

Three quantities can be visualized: **Atmospheric Concentration** (Bq/m^3), **Surface Deposition** (Bq/m^2), and **Dose Rate** (μSv/h or μGy/h depending on SIBYL settings). Each quantity is enabled or disabled using its toggle switch. Only one quantity can be displayed at a time; enabling one toggle automatically disables the others. For each quantity, clicking **LOG SCALE** toggles the color scale between logarithmic and linear, and the color scale range can be adjusted using the sliders flanking the unit label. If the output files for a quantity are not found in the I/O directory, an error message is displayed in the lower-left corner of the Map Panel when the toggle is enabled.

When **Atmospheric Concentration** is enabled, the airborne radioactivity concentration is rendered as a three-dimensional color-mapped voxel field in the Map Panel, with buildings shown in gray (**Fig. 5-1**). The **Slice** function, enabled via its toggle switch, cuts the voxel field along a plane perpendicular to a selected axis, revealing internal concentration structures that would otherwise be obscured by the outer volume rendering. Selecting **X-axis**, **Y-axis**, or **Z-axis** displays the corresponding cross-section; the position of the cutting plane is controlled by the slider below the axis buttons, with the current position in meters shown to its right. **Fig. 5-2**, **Fig. 5-3**, and **Fig. 5-4** show X-axis, Y-axis, and Z-axis slices, respectively.

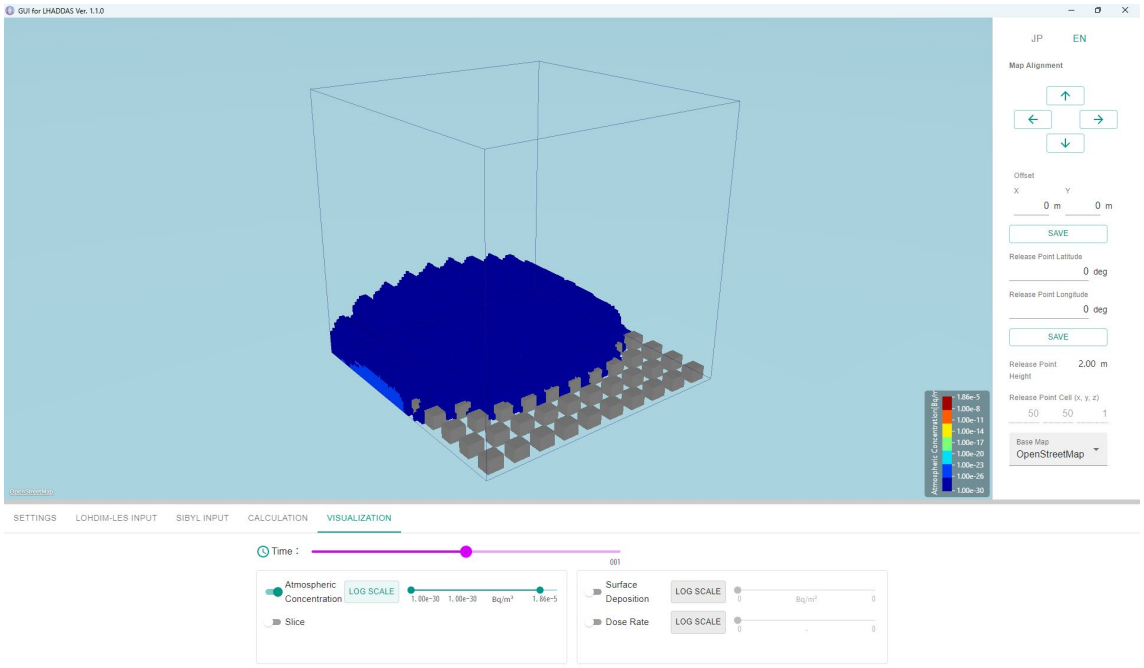


Fig. 5-1. Visualization tab showing the three-dimensional atmospheric concentration distribution rendered as a color-mapped voxel field in the Map Panel. Buildings are shown in gray.

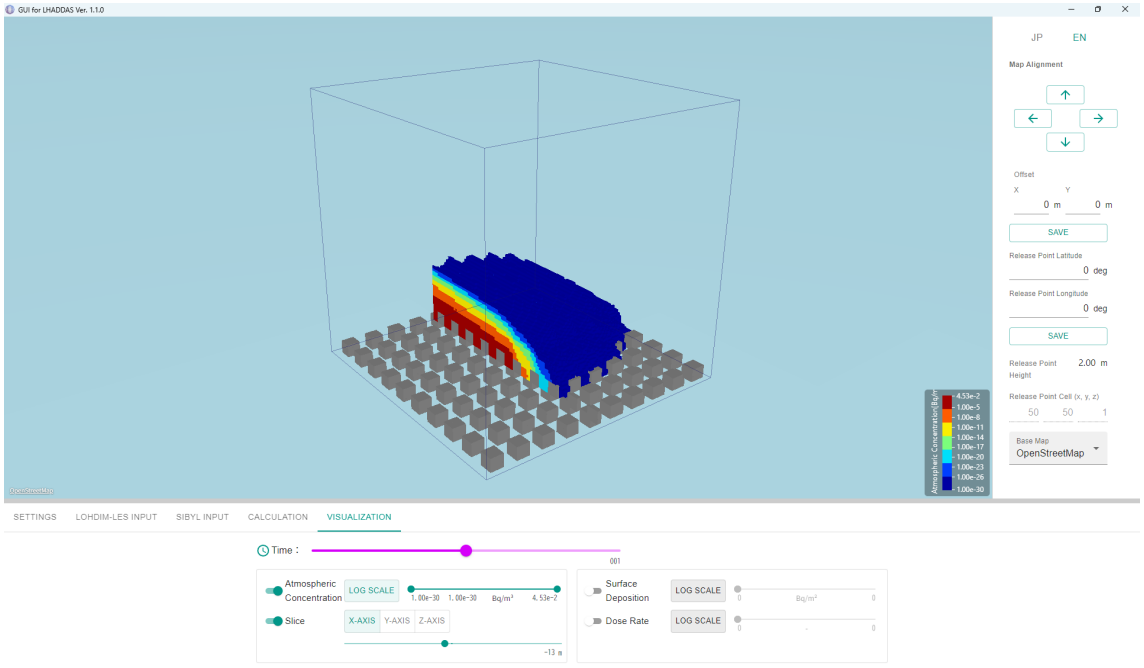


Fig. 5-2. X-axis slice of the atmospheric concentration voxel field, revealing the internal concentration structure along the X direction.

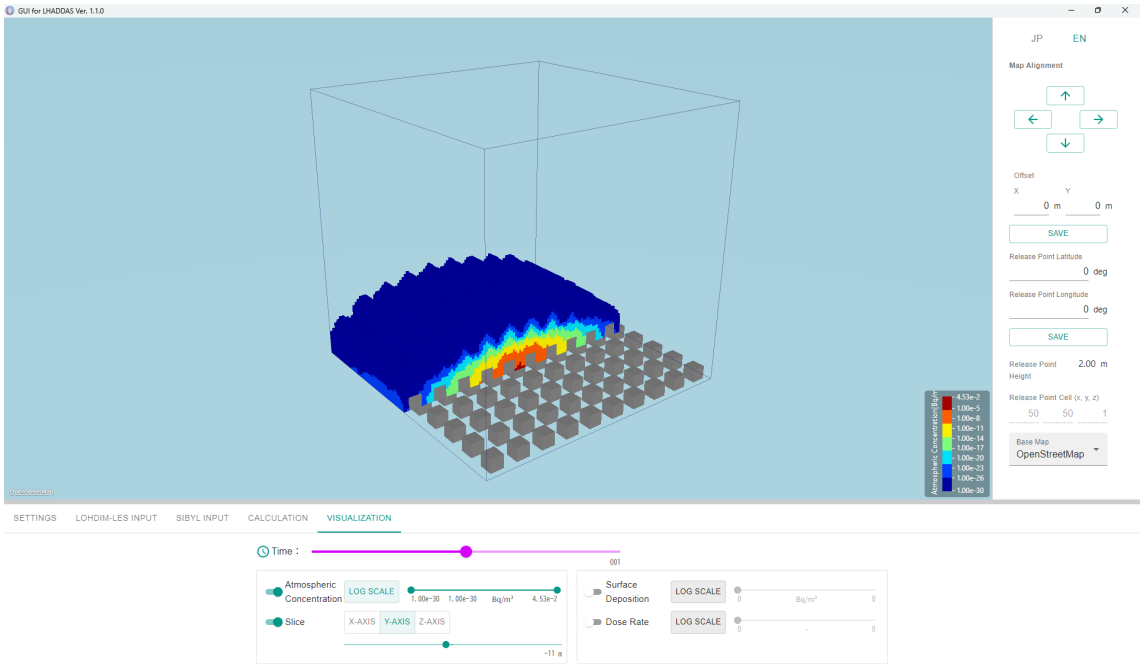


Fig. 5-3. Y-axis slice of the atmospheric concentration voxel field, revealing the internal concentration structure along the Y direction.

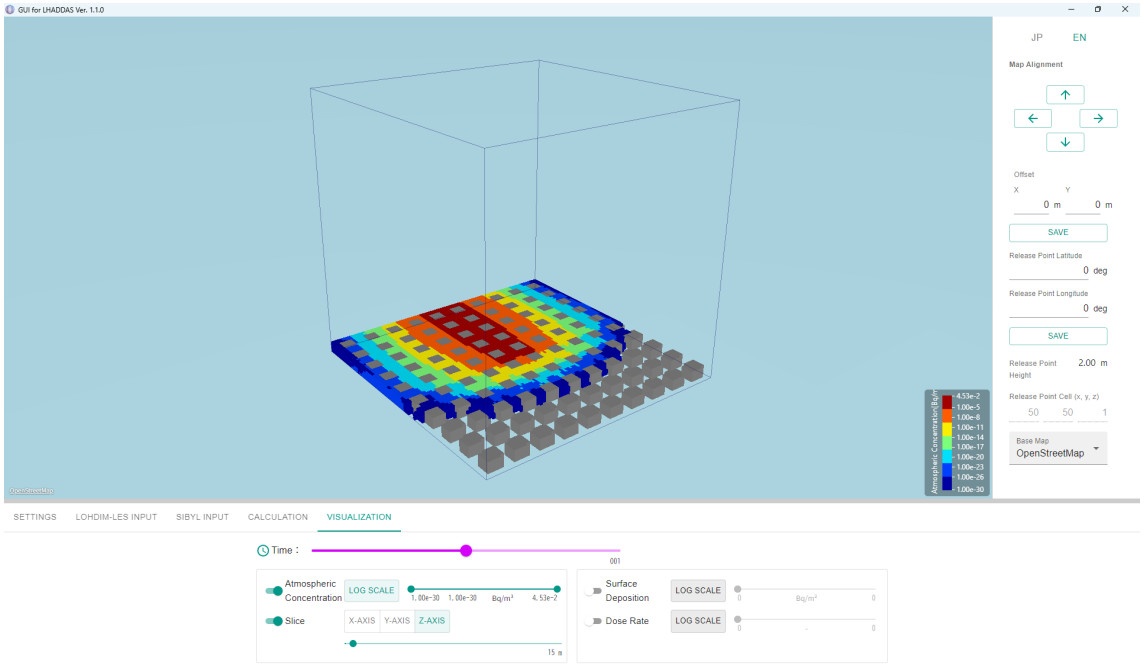


Fig. 5-4. Z-axis slice of the atmospheric concentration voxel field, revealing the internal concentration structure along the Z direction.

When **Surface Deposition** is enabled, the ground-surface deposition concentration is rendered as a two-dimensional color map on the ground plane (**Fig. 5-5**). When **Dose Rate** is enabled, the ground-level dose rate distribution is rendered similarly (**Fig. 5-6**). For both two-dimensional quantities, hovering the mouse pointer over the map displays the value at that location. Map navigation operations (pan, rotate, zoom) described in Section 4.1 remain available while any quantity is being displayed.

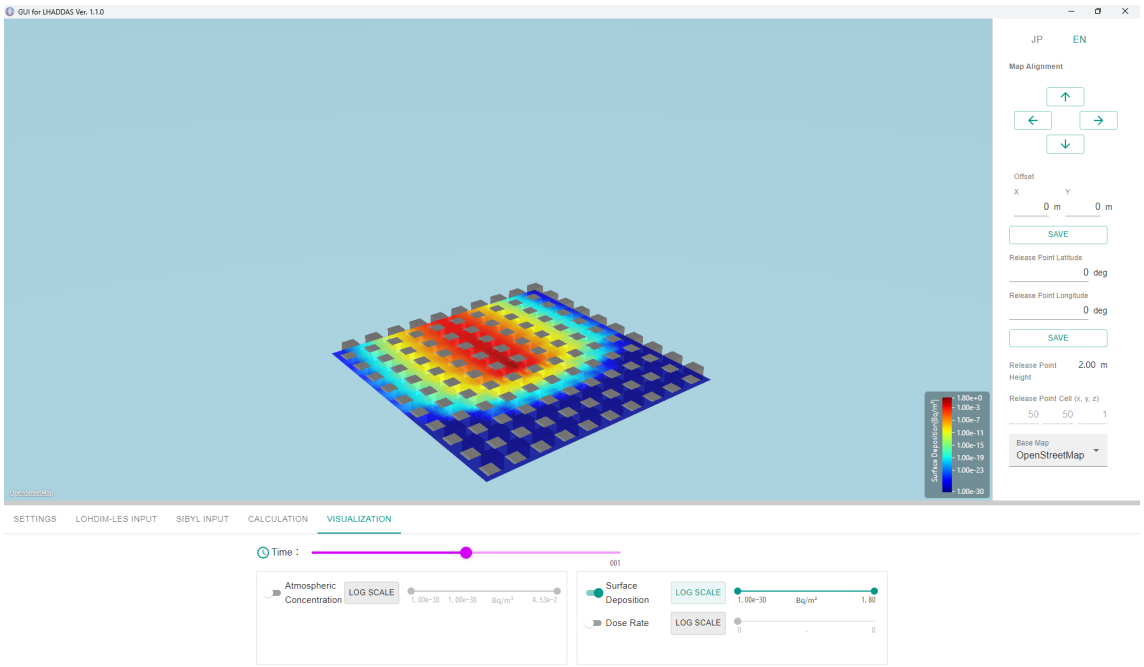


Fig. 5-5. Visualization tab showing the ground-surface deposition concentration rendered as a two-dimensional color map in the Map Panel.

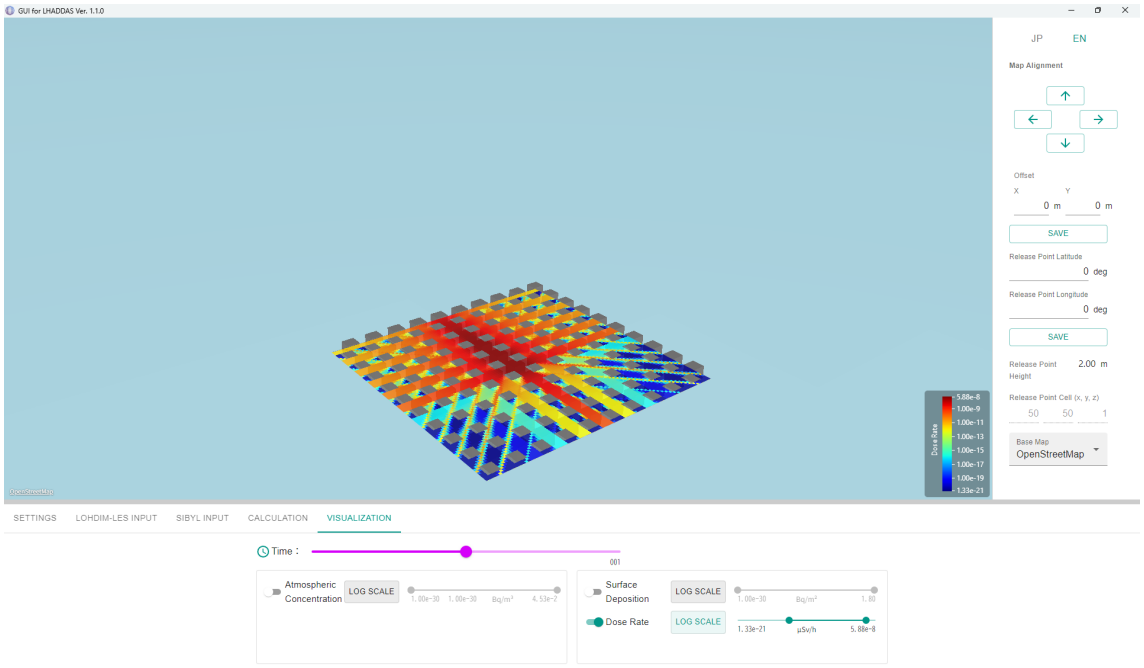


Fig. 5-6. Visualization tab showing the ground-level dose rate distribution rendered as a two-dimensional color map in the Map Panel.

6. Data Files

GUI-LHADDAS exchanges data with LOHDIM-LES and SIBYL only through files in the I/O directories. **Table 6-1** lists these files. In the file names, NNN is the index of the output time (000, 001, ...).

Table 6-1. Files exchanged between GUI-LHADDAS, LOHDIM-LES, and SIBYL.

File	Written by	Read by	Content
`LOHDIM-LES_input.data`	GUI-LHADDAS or user	LOHDIM-LES, GUI-LHADDAS	Calculation parameters (Section 5.2)
Building and vertical-grid files	User	LOHDIM-LES, GUI-LHADDAS	Input data of LOHDIM-LES, also used by GUI-LHADDAS to draw the buildings and the domain; file names are set in the LOHDIM-LES Input tab
Terrain and release-rate files	User	LOHDIM-LES	Input data of LOHDIM-LES; file names are set in the LOHDIM-LES Input tab
`SourceLocation.txt`	GUI-LHADDAS or user	GUI-LHADDAS	Latitude and longitude of the release point (Section 4.1)
`MapCorrection.txt`	GUI-LHADDAS or user	GUI-LHADDAS	Offset for map alignment (Section 4.4)
`SIBYL_input.data`	GUI-LHADDAS, LOHDIM-LES, or user	SIBYL, GUI-LHADDAS	Calculation parameters (Section 5.3)
`Z.data`	LOHDIM-LES	SIBYL, GUI-LHADDAS	Lower and upper heights of each vertical cell (m)
`SHIELD.data`	LOHDIM-LES	SIBYL	Flag of each cell: 1 for air, 9999 for building or terrain
`ELEVATION.data`	LOHDIM-LES	SIBYL	Ground elevation of each column (m)

SIBYL_PLUME_NNN.data	LOHDIM-LES	SIBYL	Air concentration averaged over the output interval (Bq/m³), 3D
SIBYL_GROUND_NNN.data	LOHDIM-LES	SIBYL	Surface deposition at the output time (Bq/m²), 2D
GUI_PLUME_NNN.data	LOHDIM-LES	GUI-LHADDAS	Air concentration for display (Bq/m³), 3D
GUI_GROUND_NNN.data	LOHDIM-LES	GUI-LHADDAS	Surface deposition for display (Bq/m²), 2D
GUI_DOSE_NNN.data	SIBYL	GUI-LHADDAS	Dose rate for display (μSv/h or μGy/h), 2D
SIBYL_RESULT_NNN.out	SIBYL	User	Dose rate in the SIBYL output format
restart.dat	LOHDIM-LES	LOHDIM-LES	Restart data for continuing a calculation

When SIBYL is run alone, the files written by LOHDIM-LES for SIBYL must be prepared by the user (Section 5.4). The parameter files and the other files written by GUI-LHADDAS can also be prepared with a text editor; GUI-LHADDAS reads them from the I/O directories (Sections 4.1, 4.4, 5.2, and 5.3).

The data files written by LOHDIM-LES and SIBYL (`Z.data`, `SHIELD.data`, `ELEVATION.data`, and the files with NNN in their names, except `SIBYL_RESULT_NNN.out`) are ASCII text files with one value per line; `Z.data` has two values per line. A three-dimensional (3D) field on a grid of nx × ny × nz cells has nx·ny·nz lines, ordered with the x index varying fastest, then the y index, and then the z index. The x index increases eastward and the y index northward from the south-west corner of the domain, and the z index increases upward from the ground. A two-dimensional (2D) field has nx·ny lines in the same order. The value for cell (i, j, k), counted from 1, is therefore on line $i + (j - 1)\cdot nx + (k - 1)\cdot nx\cdot ny$.

The files for display (`GUI_PLUME_NNN.data`, `GUI_GROUND_NNN.data`, and `GUI_DOSE_NNN.data`) contain the value −1.0 for cells inside buildings or terrain; these cells are not colored in the Map Panel. Otherwise, `GUI_PLUME_NNN.data` and `GUI_GROUND_NNN.data` contain the same values as `SIBYL_PLUME_NNN.data` and `SIBYL_GROUND_NNN.data`. Values are written in exponential format (e.g., 1.2345678E-05), so that each value takes about 17 bytes; the 3D field of a 280 × 280 × 70 grid, for example, takes about 93 MB per output time.

`SIBYL_RESULT_NNN.out` begins with a header that lists the input parameters of SIBYL, followed by one line for each cell with the x and y coordinates (m) and the dose rate. The formats

of the input files of LOHDIM-LES and SIBYL are described in Sections 5.2 and 5.3 and in the documentation of each code.

Appendix: Building from Source

This appendix describes how to build GUI-LHADDAS from source. This procedure is intended for developers. Users who wish to run GUI-LHADDAS without building from source should use the pre-built packages available on the GitHub Releases page (see Chapter 2).

Building GUI-LHADDAS requires Node.js and npm. Install Node.js by following the instructions on the official website (https://nodejs.org). After installation, verify that both tools are available by running the following commands in a terminal:

```
node -v
npm -v
```

The source code is available at https://github.com/satoh-daiki/GUI-LHADDAS. Clone or download the repository before proceeding.

A.1 Windows

Open a terminal with administrator privileges and perform the following steps.

1. Navigate to the directory containing `package.json`:

```
cd path/to/GUI-LHADDAS/app
```

2. Install the required packages:

```
npm install
```

3. Open `node_modules/vite-plugin-cesium/dist/index.mjs` and locate lines 33–37. Replace the following code:

```
if (c.root !== void 0) {
  outDir = path.join(c.root, c.build.outDir);
} else {
  outDir = c.build.outDir;
}
with:
if (c.root !== void 0) {
  // outDir = path.join(c.root, c.build.outDir);
```

```
  outDir = c.build.outDir;
} else {
  outDir = c.build.outDir;
}
```

4. Build the application:

```
npm run build:win
```

The installer `gui-lhaddas-1.1.0-win-setup.exe` is generated in the `dist` folder.

A.2 Linux

The following procedure has been verified on Ubuntu 24.04 (x64).

1. Navigate to the directory containing `package.json`:

```
cd path/to/GUI-LHADDAS/app
```

2. Install the required packages:

```
npm install
```

3. Apply the same modification to `node_modules/vite-plugin-cesium/dist/index.mjs` as described in step 3 of Section A.1.
4. Build the application:

```
npm run build:linux
```

The package `gui-lhaddas-1.1.0-linux.AppImage` is generated in the `dist` folder.

Note: GUI-LHADDAS has been verified on Ubuntu 24.04. On Red Hat-based distributions, the application may launch but the Map Panel does not function correctly due to a compatibility issue with CesiumJS.

A.3 macOS

The following procedure has been verified on macOS Sequoia 15.3 (Apple Silicon).

1. Navigate to the directory containing `package.json`:

```
cd path/to/GUI-LHADDAS/app
```

2. Install the required packages:

```
npm install
```

3. Apply the same modification to `node_modules/vite-plugin-cesium/dist/index.mjs` as described in step 3 of Section A.1.
4. Build the application:

```
npm run build:mac
```

The disk image `gui-lhaddas-1.1.0-mac.dmg` is generated in the `dist` folder. Double-click the disk image to mount it and extract the application.

Note: This procedure builds a local copy of the application for personal use. Distributing the application to other machines requires Apple notarization, which is not covered here.